ATR and MAE/MFE Stops

How to set a stop that fits each of the 58 stocks the desk trades: what each method does, how the two work together, what the current ₹550 stop does today, and what to add to the trade database.

SAMPLE	STRATEGIES	RISK UNIT	NET
81 trades · 16 sessions 21 Sep – 14 Oct 2026	breakout@4 (51) · @5 (6) breakdown@2 (24)	1R = ₹550 · 1 lot	₹18,209 = 33.1R

READ THIS FIRST The trade records hold no price path, and minute candles need a live Kite session. So every measurement here uses the **daily ATR** stored in `daily_levels`, and assumes an ATM option moves **half as much as the stock** (delta 0.5).

That is enough to show the shape of the problem. It is not enough to set the final stop. The multiplier has to come from 5-minute ATR and rebuilt MAE, which section 9 lays out.

1 The short answer

ATR and MAE/MFE are not competing choices. **ATR is the ruler a stop is measured with. MAE/MFE, measured with that ruler across all your trades, tells you where to put the mark.** On a desk that trades many stocks, MAE/MFE only works once it is converted into ATR units, so you end up needing both anyway.

> **Recommendation**
> Stop = k × ATR(14) on 5-minute bars, measured on the stock price, not the option premium. Pick k from your own winners' MAE in ATR units, pooled across every stock. Keep ₹550 as a cap on what one trade may risk, not as the thing that sets the stop.

FINDING	VALUE	WHAT IT MEANS
Room the ₹550 stop leaves, in daily ATR	`0.024 – 0.138`	A 6x spread. Some stocks get six times more breathing room than others.
Stop as a share of the stock price	`0.10% – 0.31%`	Varies about 3x, while the stocks' volatility varies about 9x.
Stopped out within 5 minutes, tightest vs widest third	`12 vs 6 of 27`	Tight room gets knocked out by ordinary noise twice as often.
Total R without each third's best trade	`−4.5 / −4.9 / +12.4`	Only the widest-room third makes money without its one big winner.
Cost of all 26 fast stop-outs	`−27.4R`	Close to the whole net result of the period (+33.1R).
How far trades that reached 4R ran on the stock	`≥ 0.55 ATR`	At least 7x the typical stop room. More room costs little next to that.

2 How each method works

The ATR stop

ATR (Average True Range) is how much a stock normally moves in one bar, averaged over the last 14 bars. "True range" counts gaps too:

```
True range = largest of ( high − low, |high − previous close|, |low − previous close| )
ATR(14) = average of the last 14 true ranges
Long stop = entry − k × ATR     Short stop = entry + k × ATR
```

The point is that each stock gets room in proportion to its own noise. A quiet stock gets a close stop, a wild one gets a wide stop, and both are equally likely to be hit by random movement.

> EXAMPLE (ILLUSTRATIVE NUMBERS)
> Two stocks, both at ₹1,000, k = 1.5. Stock A's 5-minute ATR is ₹2, so its stop sits ₹3 away, at ₹997. Stock B's ATR is ₹6, so its stop sits ₹9 away, at ₹991. A fixed ₹3 stop would suit A and be hit almost at once on B.

Trailing version. The Chandelier Exit trails k × ATR below the highest high since entry and never loosens. That gives the same volatility-scaled logic for locking profit as for the first stop.

Its weakness: k is a guess. Nothing in ATR itself says whether 1.0, 1.5 or 2.5 is right for this strategy.

MAE and MFE

For every finished trade, two numbers:

- **MAE (maximum adverse excursion):** the furthest price went against you while the trade was open.
- **MFE (maximum favourable excursion):** the furthest it went in your favour.

> EXAMPLE (ILLUSTRATIVE NUMBERS)
> Long at ₹1,000. It dips to ₹996, runs to ₹1,018, and you exit at ₹1,010. MAE = ₹4, MFE = ₹18, and you gave back ₹8 of the move.

How it sets a stop: good trades rarely go far against you before they work. Plot the MAE of every eventual winner. If 90% of winners never went more than X against you, a stop just beyond X lets the winners through and cuts most losers early.

How MFE helps: it shows how far good trades usually run, which tells you when to start trailing and how wide to trail. MFE minus the final profit is the profit you gave back.

Its weakness: it needs the price path of every trade. And in rupees it cannot be compared across stocks, which the next section deals with.

	ATR STOP	MAE / MFE
Answers	How much does this stock normally move?	How far do my winning trades go against me before they work?
Needs	Recent candles of the stock	The full price path of every past trade
Strength	Same treatment for quiet and wild stocks	Built from your own trades, not a rule of thumb
Weakness	The multiplier k is a guess	Useless across stocks unless normalised
Many different stocks?	Yes, by design	Only after dividing by ATR

3 Making MAE/MFE work across 58 stocks

Divide each trade's MAE and MFE by that stock's ATR at the moment of entry. Then every trade is in the same unit, and all 81 can go into one distribution.

MAE_{ATR} = MAE ÷ ATR at entry MFE_{ATR} = MFE ÷ ATR at entry

EXAMPLE (ILLUSTRATIVE NUMBERS)

Trade 1 went ₹4 against you on a stock with 5-minute ATR ₹1: MAE = 4.0 ATR, a very large move. Trade 2 also went ₹4 against you, on a stock with ATR ₹8: MAE = 0.5 ATR, ordinary noise. In rupees they look identical. In ATR units they are completely different trades.

Why not the units the desk already uses:

UNIT	WHAT IT FIXES	WHAT IT MISSES
Rupees	Nothing	Price level and volatility both differ by stock
% of price	Price level	Volatility. On this desk daily ATR runs from 1.5% to 13% of price.
R (₹550 ÷ lot size)	Rupee risk	R is a different distance on every stock, set by lot size. It is the problem, not the fix (section 4).
ATR units	**Price level and volatility**	**Time of day, unless 5-minute ATR is used**

The E-ratio: checking the entry on its own

The same normalisation gives a test of the entry, separate from any exit rule. Measure MFE_{ATR} and MAE_{ATR} over a fixed window after entry (say 15, 30 and 60 minutes), average each across all trades, and divide:

E-ratio = average MFE_{ATR} ÷ average MAE_{ATR}

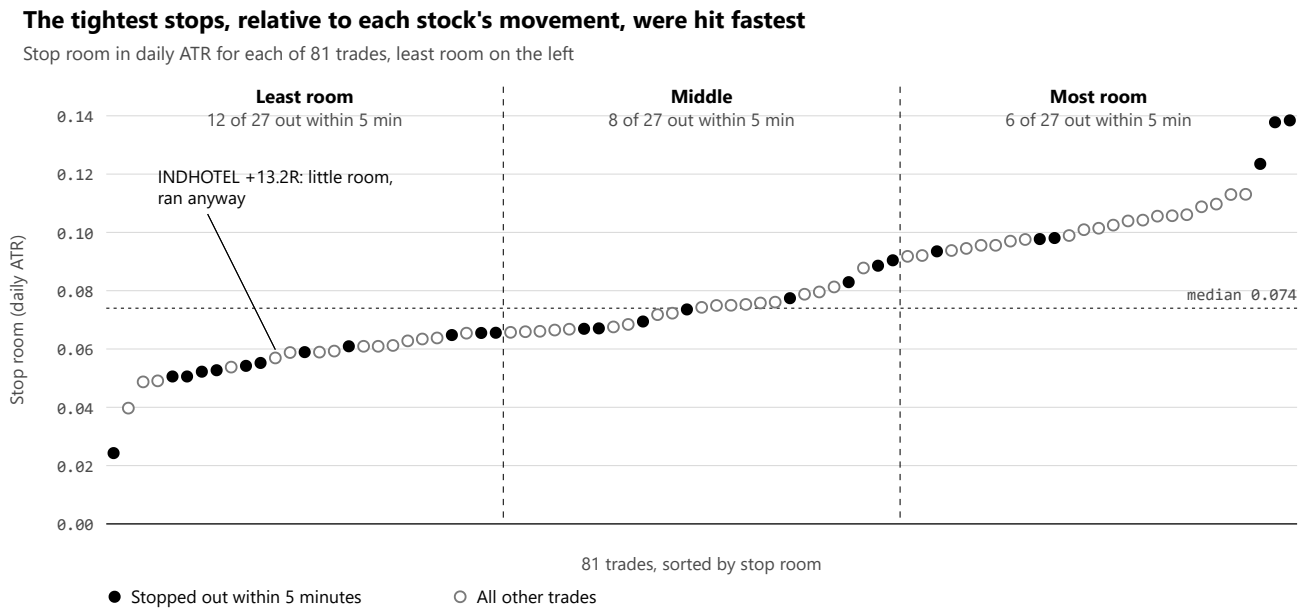
A random entry scores about 1.0. Above 1 means price moves your way more than against you after the signal. If the E-ratio is near 1, no stop or trail can rescue the strategy; if it is well above 1, the exits are where the work is.

4 What the ₹550 stop looks like against ATR today

The stop is set by the budget, not by the chart. ₹550 ÷ lot size gives the premium distance; at delta 0.5 the stock has to move twice that to hit it.

ACROSS 81 TRADES	MIN	25%	MEDIAN	75%	MAX
Stop distance, % of stock price	0.097	0.154	0.187	0.222	0.310
Stock's daily ATR, % of price	1.46	2.03	2.35	2.92	13.24
Stop room, in daily ATR	**0.024**	**0.063**	**0.074**	**0.096**	**0.138**

Why the spread exists. NSE sizes lots by contract value, so every contract is worth a broadly similar amount. That keeps ₹550 in a narrow band of the stock price: 0.10% to 0.31%, median 0.19%. The stocks' daily movement ranges far wider, 1.5% to 13%. So the ₹550 stop behaves like a percentage stop, and it is blind to volatility.



Each dot is one trade. Room = stop distance on the stock (₹550 ÷ lot size ÷ 0.5 delta) divided by the stock's 14-day daily ATR at entry. Source: Mongo strategy_trades and daily_levels.

The extremes

STOCK	STOP % PRICE	DAILY ATR %	ROOM (ATR)	LOT	RESULT R
Least room					
POLICYBZR	0.310	13.24	0.024	350	−0.86
CHOLAFIN	0.103	2.50	0.040	625	+1.31
COCHINSHIP	0.197	4.07	0.049	400	−0.33
AUROPHARMA	0.116	2.35	0.049	550	+0.65
MFSL	0.210	4.11	0.051	400	−0.29
Most room					
BRITANNIA	0.177	1.60	0.110	125	+3.94
MARUTI	0.187	1.69	0.109	50	+4.40
IRFC	0.250	2.05	0.124	5,425	0.00
GODREJCP	0.257	1.91	0.138	500	−1.05
NTPC	0.228	1.68	0.138	1,500	−1.09

POLICYBZR's 13% ATR is inflated by one stored session, 24 Sep, with a 40% high-low range. That looks like a bad print and is worth checking in `daily_levels`; its other sessions still ranged 2% to 13%. Room is not everything either: GODREJCP and NTPC had the widest stops and were still out within 3 minutes.

Longs get less room than shorts. The 57 long trades had a median room of 0.067 ATR and 20 fast stop-outs (35%). The 24 shorts had 0.092 and 6 (25%).

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5 Did the room matter?

Yes, for how often trades are knocked out early. Less clearly for total profit, because one trade dominates. Trades are split into thirds by how much room the stop had.

ROOM (DAILY ATR)	TRADES	WINNERS	REACHED 2R	OUT ≤ 5 MIN	R LOST THERE	TOTAL R	WITHOUT BEST	NET ₹
0.024 – 0.066	27	8	9	12	−13.9	+8.7	−4.5	4,792
0.066 – 0.090	27	9	8	8	−8.4	+4.9	−4.9	2,718
0.092 – 0.138	**27**	**12**	**11**	**6**	**−5.1**	**+19.5**	**+12.4**	**10,699**

- **Fast stop-outs fall steadily as room grows:** 12, then 8, then 6. Winners rise: 8, 9, 12. This is the clearest signal in the data.
- **Total R is not in order.** The tightest third beats the middle only because of INDHOTEL (+13.2R), which had little room (0.057 ATR) and ran anyway. Without it that third is −4.5R. The middle third's best is ICICIGI (+9.9R), the widest third's is BAJAJFINSV (+7.0R).
- **Only the widest third is profitable without its best trade.** It made ₹10,699, more than the other two thirds together.
- **The 26 fast stop-outs cost 27.4R between them.** Some were bad entries the stop rightly killed. The point is how many sat inside ordinary noise when they were hit.

Time of day: the case for 5-minute ATR

ENTRY TIME (IST)	TRADES	MEDIAN ROOM	OUT ≤ 5 MIN	TOTAL R
09:15 – 10:00	**19**	**0.065**	**10**	**+23.0**
10:00 – 11:00	19	0.072	2	+5.1
11:00 – 13:00	22	0.088	7	+9.2
13:00 onwards	21	0.076	7	−4.2

The first 45 minutes hold both the most fast stop-outs (10 of 19) and three of the four largest winners: INDHOTEL, ICICIGI and JSWSTEEL. That is when 5-minute bars are largest. Daily ATR is the same all day, so it cannot see this; 5-minute ATR widens the stop automatically at the open and narrows it at midday. (Entries after 13:00 stopped on 12 Oct; those 21 trades are from before.)

How far winners run

The ladder records which rung a trade reached, so the peak on the premium is known to within 1R for the 28 trades that reached 2R. Converted to the stock at delta 0.5, these are lower bounds:

PEAK REACHED	TRADES	STOCK MOVED AT LEAST (MEDIAN)	AS % OF PRICE
2R or more	28	0.28 ATR	0.73%
4R or more	12	0.55 ATR	0.93%
6R or more	5	0.74 ATR	1.64%

Against a median stop room of 0.074 ATR, a trade that reaches 4R travels at least seven times its stop distance. A stop a third wider, about 0.10 ATR, would still be less than a fifth of that move. This is the MFE side of the argument: the winners travel far, so more room at the start costs them little. The cost lands on the losers, each of which loses more rupees per stop-out, and that is what section 7 deals with.

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6 Using both together, step by step

1. **Measure the ruler.** At every entry, compute ATR(14) on the stock's 5-minute bars. `bars.ts` already keeps the closed 5-minute candles for each symbol, so this sits next to the EMA.
2. **Rebuild each trade's path.** From 1-minute candles of the stock, entry to exit, record MAE and MFE in rupees, and the minute each happened.

3. **Normalise.** Divide both by the entry ATR. Every trade, on any stock, is now in ATR units.
4. **Pick k from the winners.** Take trades that reached 2R or more. Find the MAE_{ATR} that 90% of them stayed inside. Add a small buffer for the spread. That is k.
5. **Check the losers.** Count how many losers went past k. Those would have been cut at the same loss or smaller. Losers that never reached k would have lasted longer; check what they did next.
6. **Set the trail from MFE.** Use winners' MFE_{ATR} to decide when trailing starts and how far behind it sits, e.g. a Chandelier trail of m × ATR below the best price since entry.
7. **Test it on trades it was not fitted to.** Choose k on the first half of the trades by date, then check it on the second half. If it only works on the half it was chosen from, it is fitted noise.
8. **Release it as a new version.** Versions are immutable, so the ATR stop becomes long v6 and short v4, and every trade stays labelled with the rule that produced it.

> WORKED EXAMPLE (ILLUSTRATIVE NUMBERS)
> Long on a ₹1,500 stock. 5-minute ATR at entry is ₹4.00 and k = 1.2, so the stop is ₹4.80 below entry, at ₹1,495.20 on the stock. The engine watches the stock's price from the tick feed it already has and exits the option when the stock trades at ₹1,495.20. At delta 0.5 the premium should be about ₹2.40 lower by then. With a lot of 250, that is about ₹600 of risk, a little over the ₹550 budget. With a lot of 1,000 it would be ₹2,400. Section 7 is about what to do with that difference.

Why the stop should sit on the stock, not the premium. The premium moves with delta, time decay and volatility at once. A premium stop can fire because volatility dropped, even when the stock did exactly what the signal predicted. A stop on the stock measures only what the strategy actually forecasts. It also costs nothing extra: the stock's price is already streaming for every symbol.

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7 Position sizing: fixed rupees or fixed room

You can hold one of the two constant, not both. Today the rupee risk is fixed and the room varies sixfold. With an ATR stop the room is fixed and the rupee risk varies:

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Risk per lot = k × ATR × delta × lot size
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If every trade in the sample had been given the median room (0.074 ATR), one lot would have risked between ₹298 and ₹1,696, with a median of ₹555.

A gentler option is a floor: keep ₹550 wherever it already gives enough room, and widen only the tight stops.

MINIMUM ROOM	TRADES WIDENED	MEDIAN RISK	90TH PCT	HIGHEST	OVER ₹825
0.066 ATR	29	₹550	₹674	₹1,493	2
0.075 ATR	**42**	**₹555**	**₹766**	**₹1,696**	**4**
0.092 ATR	55	₹681	₹940	₹2,081	20

The 0.092 row brings every trade up to the widest third's room, but a quarter of trades would then risk more than 1.5R.

Three ways to handle the lots

RULE	HOW IT WORKS	COST
Cap and skip	Take 1 lot. If 1 lot risks more than a cap (say ₹825, 1.5R), skip the trade.	Misses some volatile stocks; risk per trade varies up to the cap.
Scale lots	lots = largest whole number with risk ≤ ₹550, minimum 1. Quiet stocks with small lots get 2 or 3 lots.	Puts more capital in quiet stocks; still needs a cap for the 1-lot case.
Filter only	Keep the ₹550 stop; skip any setup where ₹550 leaves less than the minimum room.	Would have skipped INDHOTEL (+13.2R). On this sample, filtering the tightest third would have cost 8.7R.

> **Suggested**
> Scale lots plus a cap: lots = max(1, floor(₹550 ÷ risk per lot)), and skip when one lot exceeds ₹825. R then means each trade's actual risk, so results stay comparable in R. This is your decision; the numbers above are what each choice would have done.

8 What to add to the trade database

None of these change the trading rule, so none need a version bump. They make the next study answerable instead of estimated.

FIELD / COLLECTION	HOLDS	WHY	BACK-FILL?
`entryAtr` on the trade	ATR(14) on 5-min and daily bars at entry, in rupees and % of price	The unit every MAE/MFE is divided by. Must be the value known at entry, not today's.	Yes, from candles
`excursion.stock` on the trade	Running MAE, MFE and the time of each, on the stock price	The core of the method. Updated on the existing 2-second tracking loop from prices already streaming.	Yes: stock minute candles do not expire
`excursion.premium` on the trade	Running MAE, MFE and times on the option premium	Shows when the premium and the stock disagree (volatility, decay).	Only while the contract is still listed
`instrument.instrumentToken`, `underlyingToken`	Kite tokens of the option and the stock	The exit study lost 28 trades because expired option tokens vanish from the instrument list.	Stock: yes. Option: only unexpired.
`stopBasis`, `stopUnderlying`	"premium" or "stock-atr", and the stock price the stop sits at	Lets one blotter hold both rules side by side without mixing them up.	Not needed
`deltaAtEntry`	Estimated delta from premium, strike, spot and expiry	Replaces the 0.5 assumption used throughout this document.	Yes, approximately
`trade_paths` new collection	One document per trade: stock and premium samples (every 10 s or each minute) from entry to exit	Any future stop or trail can be replayed exactly, not estimated. Write it to the disk journal first, like trades.	Stock side: yes
`shadowExits` on the trade	Where candidate rules (e.g. 1.0, 1.5, 2.0 × ATR) would have exited, and at what R	Compares rules live, on the same entries, without changing what is traded.	From `trade_paths`

FIELD / COLLECTION	HOLDS	WHY	BACK-FILL?
`atr14` on `daily_levels`	Daily ATR per stock per session, from the nightly scan	Ready at the open for the daily view and for sizing, no recomputation.	Yes, from stored OHLC

The big one is back-filling the stock side. All 81 trades already store the stock, side, entry and exit times and the entry price. Stock minute candles stay available long after the options expire. So MAE/MFE on the stock can be rebuilt for every past trade today, without waiting weeks for new data. It takes about 81 historical calls, roughly 30 seconds under Kite's 3-per-second limit, and needs a signed-in session.

9 Plan, in order

1. **Back-fill** 5-minute ATR and stock MAE/MFE for all 81 trades from minute candles. Needs a live Kite session.
2. **Add the logging fields** in section 8, so premium MAE/MFE and paths are captured from the next session onward. No version bump.
3. **Choose k** from winners' pooled MAE_{ATR} on the first half of trades; check it on the second half. Compute the E-ratio at 15, 30 and 60 minutes.
4. **Decide the sizing rule** from section 7.
5. **Run the ATR stop in shadow** for two to three weeks beside the live ₹550 stop, on the same entries.
6. **Release** as long v6 and short v4 if the shadow results hold up on trades it was not fitted to.

10 Limits of this analysis

- **Daily ATR, not 5-minute.** It ranks stocks by volatility correctly but cannot see time of day. The final k must use 5-minute ATR.
- **Delta 0.5 is assumed.** Real ATM delta drifts roughly between 0.4 and 0.6, so each room figure is approximate.
- **Small sample, concentrated profit.** 81 trades over 16 sessions, 27 per group. INDHOTEL alone is 40% of the period's net R, and moving one trade between groups changes the totals.
- **Premium stop, stock measurement.** The live stop is on the premium, which can fire on a volatility drop with no stock move. Comparing it with stock ATR is an approximation.
- **MFE figures are lower bounds,** available only for the 28 trades that reached 2R.
- **INDHOTEL's ATR uses only 4 sessions,** because its trade on 21 Sep came a few sessions after stored history begins.
- **Paper fills.** All figures are simulated entries and exits.

11 Further reading

1. **How to Measure the Edge of Your Trading Entry Signals** (IASG, 2015): the E-ratio method from Curtis Faith's *Way of the Turtle*, MAE and MFE divided by ATR so many markets pool into one test.
   `iasg.com/?p=596`
2. **Maximum Adverse Excursion**, John Sweeney (Wiley, 1996): the original method for setting stops from MAE distributions.
   `nexusfi.com/wiki/trading-wiki/Maximum-Adverse-Excursion`

3. **MAE / MFE distributions**: a short practical guide to plotting both excursions to tune stops and targets together.
   `luxalgo.com/library/concept/mae-mfe-distributions.md`
4. **Chandelier Exit**, Chuck LeBeau: trailing k × ATR below the highest high, ratcheted.
   `barchart.com/education/technical-indicators/chandelier-exit`
5. **The Probability of Backtest Overfitting**, Bailey, Borwein, López de Prado and Zhu (2016): how to check that a tuned k is real before trusting it.
   `papers.ssrn.com/abstract=2326253`

Paper fills. This desk places no orders. Data: Mongo `strategy_trades` (closed trades on the versions with the order-book check) and `daily_levels` (session high, low and close), read 14 Oct 2026.